

Higher Green

G finite abstract simplicial complex.

$$\chi_m(G) = \sum_{\mathbf{x} \in G^m, \bigcap \mathbf{x} \in G} \omega(\mathbf{x})$$

m-th characteristic. Total m-energy

$$g_m(\mathbf{x}) = \omega(\mathbf{x}) \chi_m(U(\mathbf{x}))$$

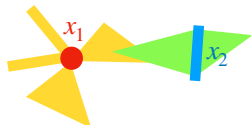
k point Green function

energy of a k-particle configuration.

Theorem: for all G, for all k and m

$$\chi_m(G) = \sum_{\mathbf{x} \in G^k} g_m(\mathbf{x})$$

"Total m-energy of all k-particle configurations is the m-energy of G"



$$g_m(x_1, x_2) = \omega(x_1) \omega(x_2) \chi_m(U(x_1) \cap U(x_2))$$

Notation

G finite abstract simplicial complex.

$$\mathbf{x} = (x_1, \dots, x_k) \quad \mathbf{x} = (x_1, \dots, x_m)$$

k tuple of particles in G m- molecule if $\bigcap x \neq \emptyset$

(order of particles do not matter)

$$\omega(\mathbf{x}) = \omega(x_1) \cdots \omega(x_m)$$

energy of a tuple of simplices

$$\bigcap \mathbf{x} = \bigcap_{j=1}^m x_j$$

simultaneous intersection

$$U(\mathbf{x}) = \bigcap_{j=1}^k U(x_j)$$

star of a k-particle configuration

m energy level

k number of particles

These are independent numbers

The story is Bosonic and force related

Case m=1:

1-point functions :
"vacuum expectations"

$$\chi(G) = \sum_{x \in G} \omega(x)$$

Euler Characteristic

$$g(x) = \omega(x) \chi(U(x))$$

curvature or index, dual to $\omega(x)$

$$\chi(G) = \sum_{x \in G} \omega(x) \chi(U(x))$$

a Gauss Bonnet result

super trace of 2-point Green function. Using the valuation property and the definition of $\chi(G)$, this gives:

$$0 = \sum_{x \in G} \omega(x) \chi(S(x))$$

sphere theorem

Case m=2:

2-point functions $g(x, y)$
"potential energy"
"probability amplitude"

$$\chi_2(G) = \sum_{x, y \in G, x \cap y \in G} \omega(x) \omega(y)$$

Wu Characteristic

$$g(x, y) = \omega(x) \omega(y) \chi(U(x) \cap U(y))$$

Green functions

$$\chi_2(G) = \sum_{x, y \in G} g(x, y)$$

energy theorem

$$0 = \sum_{x, y \in G} \omega(x) \omega(y) \chi(S(x) \cap S(y))$$

quadratic sphere theorem

Proof:

generalize!

The result works for any multi-linear valuation.
energize the simplices arbitrarily

$$\chi_m(G) = \sum_{\mathbf{x} \in G^m, \bigcap \mathbf{x} \in G} h(x_1, \dots, x_m)$$

The map $h \rightarrow g_k(h)$ is linear.

Verify for basis.

Local Valuation

$$\chi_m(B(x)) = \chi_m(U(x)) - (-1)^m \chi_m(S(x))$$

Sphere formula

For all positive k and m and all G

$$\sum_{\mathbf{x} \in G^k} \omega(\mathbf{x}) \chi_m(S(\mathbf{x})) = 0$$

Manifolds

For all q-manifolds with boundary

$$\chi_m(G) = \chi(G) - (-1)^m \chi(\delta G)$$

it can be seen as a generalization of

$$\chi_m(B(x)) = \chi_m(U(x)) - (-1)^m \chi_m(S(x))$$

using that Barycentric refinement reserves higher characteristic:

$$\chi_m(G) = \chi_m(G_1)$$

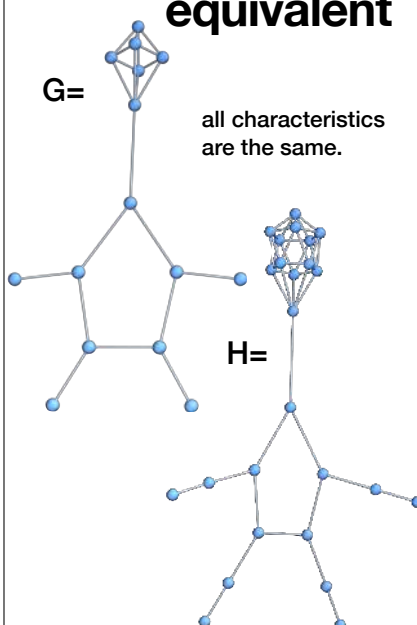
For even dimensional manifolds:

$$\chi_m(G) = \chi(G)$$

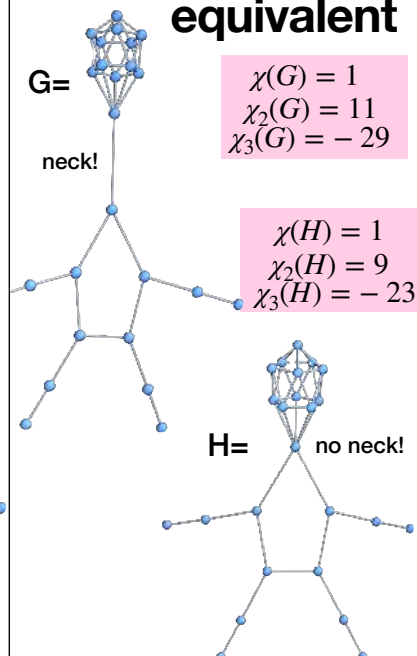
For odd-dimensional manifolds:

$$\chi_m(G) = -(-1)^m \chi(\delta G)/2$$

Topological equivalent



Not topological equivalent



$$\begin{aligned} \chi(G) &= 1 \\ \chi_2(G) &= 11 \\ \chi_3(G) &= -29 \end{aligned}$$

$$\begin{aligned} \chi(H) &= 1 \\ \chi_2(H) &= 9 \\ \chi_3(H) &= -23 \end{aligned}$$

This is the end of book I of

ELEMENTS OF FINITE GEOMETRY

WITH COLOURED DIAGRAMS AND SYMBOLS
ART MEETS SCIENCE EDITION

BY OLIVER KNILL

SURVEYOR OF IMAGINARY MATHEMATICAL ISLANDS
AND AUTHOR OF NUMEROUS MATHEMATICAL WORKS

